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Master Thesis

Beyond $\alpha = 0.05$
A Data Driven System for Alpha Level Selection.

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Abstract

When researchers run a statistical test, they must decide how much evidence is required to call a result statistically significant. This decision is controlled by the significance level, denoted by α. For decades, nearly all researchers have used the same threshold: $\alpha = 0.05$. This implies a 5% chance of incorrectly concluding that an effect exists when it actually does not. The problem is that this threshold was never selected based on strong scientific justification; instead, it gradually became a convention.

Using the same significance level for every study, regardless of sample size or the expected strength of an effect, has created practical problems. Many findings that were originally declared significant have later failed to replicate when independent researchers attempted to reproduce them. This phenomenon is commonly referred to as the replication crisis and affects multiple disciplines, including psychology, medicine, and the social sciences. Some researchers have suggested lowering the standard threshold, for example to $\alpha = 0.005$, but such proposals still rely on a single fixed number applied uniformly to all studies.

This thesis proposes a different approach. Instead of using one universal threshold, it introduces a system that recommends a study specific significance level. The system operates in two stages. First, a machine learning model is trained on data from 664 real studies that were independently replicated. The model learns how appropriate significance levels tend to vary depending on the sample size and the expected effect size of a study. Based on these inputs, the model generates a personalized baseline value for α.

In the second stage, this baseline value can be adjusted according to the context of the study. Researchers who want to be more cautious about false positive findings can choose a stricter threshold. Adjustments can also be made for one tailed tests or when multiple hypotheses are tested simultaneously. These adjustments follow transparent rules so that the reasoning behind the final recommended value remains clear.

The machine learning model was evaluated using previously unseen data and achieved a mean absolute prediction error of approximately 0.014, which is small relative to the typical range of α values between 0.001 and 0.15. The complete system was then tested on 20 real studies that had

originally been reported as significant using $\alpha = 0.05$ but later failed to replicate. In 11 of these cases, the system would have recommended a stricter threshold, meaning that the original results might not have been declared significant.

The main conclusion of this work is that treating $\alpha = 0.05$ as a universal rule is not always appropriate. Different studies have different characteristics, and the significance level should reflect these differences. The proposed system provides researchers with a practical tool for making more informed decisions about significance thresholds while maintaining transparency and interpretability.

Zusammenfassung

Wenn Forschende einen statistischen Test durchführen, müssen sie entscheiden, wie stark die Evidenz sein muss, damit ein Ergebnis als statistisch signifikant gilt. Dies wird durch das Signifikanzniveau bestimmt, das mit α bezeichnet wird. Seit Jahrzehnten verwenden nahezu alle Forschenden denselben Schwellenwert: $\alpha = 0{,}05$. Dies bedeutet, dass eine Wahrscheinlichkeit von 5 % besteht, fälschlicherweise zu schließen, dass ein Effekt existiert, obwohl dies tatsächlich nicht der Fall ist. Problematisch ist jedoch, dass dieser Wert nie auf einer klaren wissenschaftlichen Begründung beruhte, sondern sich im Laufe der Zeit als Konvention etabliert hat.

Die Verwendung desselben Schwellenwerts für jede Studie – unabhängig von der Stichprobengröße oder der erwarteten Effektstärke – hat jedoch praktische Probleme verursacht. Viele Ergebnisse, die ursprünglich als signifikant berichtet wurden, konnten später nicht repliziert werden, wenn andere Forschende versuchten, die Studien zu wiederholen. Dieses Phänomen wird als Replikationskrise bezeichnet und betrifft verschiedene wissenschaftliche Bereiche, darunter Psychologie, Medizin und Sozialwissenschaften. Einige Forschende haben vorgeschlagen, den Standardwert zu senken, beispielsweise auf $\alpha = 0{,}005$. Solche Vorschläge beruhen jedoch weiterhin auf einem festen Schwellenwert, der für alle Studien gleichermaßen gilt.

Diese Arbeit verfolgt daher einen anderen Ansatz. Anstatt einen einheitlichen Wert für alle Studien zu verwenden, wird ein System entwickelt, das ein studienspezifisches Signifikanzniveau empfiehlt. Das System arbeitet in zwei Schritten. Zunächst wird ein Machine-Learning-Modell mit Daten aus 664 realen Studien trainiert, die unabhängig repliziert wurden. Das Modell lernt, wie sich geeignete Signifikanzniveaus in Abhängigkeit von der Stichprobengröße und der erwarteten Effektgröße einer Studie verändern. Auf dieser Grundlage wird ein personalisierter Ausgangswert für α berechnet.

Im zweiten Schritt kann dieser Ausgangswert an den Kontext der Studie angepasst werden. Forschende, die besonders vorsichtig gegenüber falschpositiven Ergebnissen sein möchten, können beispielsweise einen strengeren Schwellenwert wählen. Weitere Anpassungen sind möglich, etwa bei einseitigen Tests oder wenn mehrere Hypothesen gleichzeitig geprüft wer-

den. Diese Anpassungen folgen klaren und transparenten Regeln, sodass nachvollziehbar bleibt, wie der endgültige empfohlene Wert entsteht.

Das Machine-Learning-Modell wurde anschließend mit Daten getestet, die während des Trainings nicht verwendet wurden. Dabei ergab sich ein mittlerer absoluter Vorhersagefehler von etwa 0,014, was im Vergleich zum typischen Bereich von α-Werten zwischen 0,001 und 0,15 gering ist. Zusätzlich wurde das vollständige System an 20 realen Studien getestet, die ursprünglich bei $\alpha = 0{,}05$ als signifikant berichtet wurden, später jedoch nicht repliziert werden konnten. In 11 dieser Fälle hätte das System einen strengeren Schwellenwert empfohlen, sodass die ursprünglichen Ergebnisse möglicherweise gar nicht als signifikant eingestuft worden wären.

Die zentrale Erkenntnis dieser Arbeit ist, dass die pauschale Verwendung von $\alpha = 0{,}05$ nicht immer sinnvoll ist. Unterschiedliche Studien besitzen unterschiedliche Eigenschaften, und das Signifikanzniveau sollte diese Unterschiede berücksichtigen. Das entwickelte System bietet Forschenden ein praktisches Werkzeug, um fundiertere Entscheidungen über geeignete Signifikanzschwellen zu treffen.

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Symbols, Constants and Abbreviations

Greek Letters and Statistical Symbols

α Significance level (probability of a Type I error)

$\alpha_{\textbf{base}}$ Baseline significance level predicted by the machine learning model

$\alpha_{\textbf{final}}$ Final recommended significance level after context adjustments

$\alpha_{\textbf{opt}}$ Optimal significance level derived from replication data

β Probability of a Type II error

η^2 Eta-squared; effect size measure for F-tests

ϕ Phi coefficient; effect size for chi-squared tests

μ Population mean

σ Population standard deviation

χ^2 Chi-squared test statistic

Roman Letters and Mathematical Notation

A Candidate set of significance levels used for optimal alpha selection

d Standardised effect size (Cohen's d)

df Degrees of freedom

df_1 Numerator degrees of freedom (F-test)

df_2 Denominator degrees of freedom (F-test)

F F-test statistic

H_0 Null hypothesis

H_1 Alternative hypothesis

k_1 Risk tolerance adjustment factor

k_2 Test directionality adjustment factor

k_3 Multiple testing correction factor

L Expected loss function

m Number of simultaneous hypothesis tests

N Sample size

n Number of observations in the test set

p p-value

$p_{\textbf{orig}}$ p-value reported in the original study

r Pearson correlation coefficient (effect size)

t t-test statistic

T General test statistic

$t_{\textbf{obs}}$ Observed test statistic value

Z Standardised test statistic (z-score)

Constants and Fixed Thresholds

0.05 Conventional significance level

0.005 Proposed stricter significance threshold

0.001 Lower bound applied to α_{final}

0.15 Upper bound applied to α_{final}

42 Random seed used for reproducible model training

200 Number of decision trees in the Random Forest model

0.80 / 0.20 Training / test split ratio used during model evaluation

Abbreviations

ASA American Statistical Association

FReD FORRT Replication Database

FORRT Framework for Open and Reproducible Research Training

LOOPR Life Outcomes of Personality Replication project

MAE Mean Absolute Error

ML Machine Learning

MSE Mean Squared Error

NHST Null Hypothesis Significance Testing

OLS Ordinary Least Squares

OSF Open Science Framework

RF Random Forest

RMSE Root Mean Squared Error

UI User Interface

XGBoost Extreme Gradient Boosting algorithm

Software and Libraries

NumPy Python library for numerical array operations

pandas Python library for tabular data manipulation

pickle Python library for object serialisation

scikit-learn Python machine learning library

Streamlit Framework for building interactive data applications

GitHub Web based version control and repository hosting platform

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Part I.

Introduction and Problem Setup

1. Introduction

1.1. Background and Motivation

Statistical hypothesis testing is one of the most commonly used methods in scientific research. In empirical research, researchers aim to assess whether an observed effect represents a real relationship or whether it may have occurred due to random variation.

To answer this question, researchers define two competing statements. The first is the null hypothesis, denoted by H_0, which usually represents "no effect" or "no difference". The second is the alternative hypothesis, denoted by H_1, which represents the presence of an effect.

After collecting data, researchers calculate a value known as the *p-value*. [Fis25] introduced the p-value as a way to measure how compatible the observed data are with the null hypothesis. In simple terms, the p-value answers the following question:

> If the null hypothesis were true, how likely would it be to observe data like the ones we obtained?

The *p-value* is the probability of observing a test statistic at least as extreme as the observed value, assuming that the null hypothesis is true. It measures how compatible the observed data are with the null hypothesis.

More formally, if T is a test statistic and t_{obs} is its observed value, the p-value can be written as

$$p = P(T \geq t_{\text{obs}} \mid H_0) \tag{1.1}$$

The significance level α represents the probability of committing a Type I error, meaning that the null hypothesis is rejected even though it is true. Formally,

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true}) \tag{1.2}$$

To make a decision, the p-value is compared to a pre chosen threshold called the significance level, denoted by α. According to the framework

developed by [NP33], the null hypothesis is rejected if

$$p \leq \alpha \tag{1.3}$$

The significance level α represents the probability of making a Type I error, which means rejecting the null hypothesis even though it is true [NP33]. Formally,

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true}) \tag{1.4}$$

In practice, researchers often choose $\alpha = 0.05$. Other common values are 0.10 and 0.01. The value 0.05 has become the standard in many scientific fields. However, [Fis25] originally suggested 0.05 as a convenient reference point, not as a universal rule. Later research has shown that the dominance of 0.05 is mainly historical and conventional rather than theoretically optimal [CD82].

At first glance, this decision rule seems clear and simple. If the p-value is smaller than 0.05, the result is called "statistically significant". If it is larger, it is not.

However, problems arise when the p-value is very close to the threshold. For example, if $p = 0.049$, the null hypothesis is rejected. If $p = 0.051$, it is not rejected. The numerical difference is very small, but the conclusion changes completely. In practice, researchers sometimes describe such results as "almost significant" or "marginally significant", even though hypothesis testing formally allows only two decisions.

This discomfort has become more serious in recent years. Large replication projects have shown that many statistically significant findings cannot be reproduced in independent studies. For example, the Open Science Collaboration [Ope15] reported that a substantial proportion of findings in psychology failed to replicate . In addition, the American Statistical Association warned that p-values should not be used mechanically or interpreted without context [WL16].

These developments suggest that the routine use of a fixed significance level, especially $\alpha = 0.05$, may not always be appropriate. More recent work confirms that replication failure remains a persistent challenge across disciplines and that the validity of initial findings is systematically overestimated when significance thresholds are applied uniformly [Nos+22].

1.2. Research Problem

Even though many researchers are aware of the limitations of fixed significance levels, the value $\alpha = 0.05$ is still used almost automatically in most empirical studies. This creates several concerns.

First, studies differ in important ways. Some studies have small sample sizes, while others include thousands of observations. Some studies aim to detect very small effects, while others focus on large effects. Despite these differences, the same significance level is often applied.

Additionally, fixing α without accounting for sample size or expected effect size can create an imbalance between false positive and false negative error rates, a relationship explored in detail in Chapter 2.

Power depends on three main factors: the significance level α, the sample size, and the true effect size. [Coh88] emphasized that these quantities are closely related and should be considered together when planning a study. Fixing α without considering sample size or expected effect size may therefore lead to an imbalance between false positive and false negative decisions.

Third, the strict division between "significant" and "not significant" can create misleading interpretations when p-values are close to the threshold. Two results that are almost identical numerically can lead to completely different conclusions simply because one falls slightly below 0.05 and the other slightly above.

Although the statistical literature contains proposals to modify or lower the conventional threshold, such as recommending $\alpha = 0.005$ for certain claims [Ben+18], these proposals still rely on fixed values. There is currently no widely used method that systematically adapts the significance level to the characteristics of a specific study.

This gap forms the starting point of the present thesis.

1.3. Research Objective

The main objective of this thesis is to develop and evaluate a systematic approach for choosing the significance level α. Instead of treating α as a fixed constant, this thesis proposes to view it as a flexible value that can depend on study characteristics. In particular, the proposed approach uses information from past replication studies to inform the choice of α. Study characteristics such as sample size and effect size are used to generate a data based baseline recommendation.

In a second step, this baseline value can be adjusted based on contextual factors, such as the researcher's tolerance for risk or the number of hypotheses being tested.

The goal is not to replace statistical reasoning or to claim that there is one universally correct significance level. Instead, the goal is to provide a structured and transparent recommendation that helps researchers make more informed decisions, especially in situations where the p-value is close to the threshold.

The central research question of this thesis is:

> *Can a data driven and context sensitive approach improve the choice of the significance level compared to the conventional fixed value of $\alpha = 0.05$?*

1.4. Structure of the Thesis

The remainder of this thesis is organized as follows:

- Chapter 2 introduces the historical foundations of statistical hypothesis testing. It explains how significance testing developed and discusses the main ideas of Fisher and the Neyman–Pearson framework. It also describes how the commonly used significance level $\alpha = 0.05$ became widely adopted in scientific research.
- Chapter 3 reviews previous research on how to choose a significance level. It discusses several approaches from the literature, including fixed thresholds, power based methods, and decision theoretic approaches. The limitations of these methods are also highlighted.
- Chapter 4 explains the development of the proposed Alpha Recommendation System. It describes how the dataset is constructed, how optimal alpha values are defined using replication studies, and how a machine learning model is used to predict a a α_{base}. The chapter also explains the context based adjustment step that modifies theα_{base} based on study characteristics.
- Chapter 5 presents the deployment of the system. It explains how the trained machine learning model is integrated into an interactive application using Streamlit and how users can interact with the system to obtain an alpha recommendation.
- Chapter 6 evaluates the performance of the proposed system. The recommended alpha values are tested using selected studies and compared with decisions based on the conventional significance level.

- Chapter 7 concludes the thesis by summarizing the main findings, discussing the limitations of the approach, and suggesting possible directions for future research.

Part II.

Historical Foundations of Statistical Hypothesis Testing

2. Origins of Modern Hypothesis Testing

2.1. Fisher's Approach to Significance Testing

The modern practice of statistical hypothesis testing has its roots in the early twentieth century. One of the central figures in its development was Ronald A. Fisher. In his book *Statistical Methods for Research Workers*, introduced the idea of testing a null hypothesis using probability calculations [Fis25].

Fisher's main contribution was the introduction of the p-value as a measure of evidence against the null hypothesis. In his framework, the null hypothesis represents a default assumption, usually that there is no effect or no difference. After collecting data, a test statistic is calculated. The p-value then measures how unusual the observed result would be if the null hypothesis were true.

Importantly, Fisher did not originally present hypothesis testing as a strict decision rule. Instead, he treated the p-value as a continuous measure of evidence. A smaller p-value indicated stronger evidence against the null hypothesis. For example, a p-value of 0.01 was considered stronger evidence than a p-value of 0.04.

Fisher suggested conventional thresholds such as 0.05 and 0.01 as useful reference values [Fis25]. However, these thresholds were not meant to be rigid boundaries. They were intended as practical guidelines. Later historical analysis shows that the choice of 0.05 was influenced by convenience and available statistical tables rather than by a formal optimality argument [CD82].

In Fisher's framework, there was no formal definition of Type I and Type II errors, and no explicit requirement to fix the significance level in advance. The p-value itself was interpreted as a measure of how surprising the data were under the null hypothesis.

Thus, Fisher's approach focused mainly on measuring evidence rather than making long run error controlled decisions.

2.2. The Neyman-Pearson Framework

While Fisher focused on the interpretation of evidence, Jerzy Neyman and Egon Pearson developed a different and more formal framework for hypothesis testing. In their influential 1933 paper, they introduced a decision theoretic approach to testing statistical hypotheses [NP33].

In the Neyman-Pearson framework, hypothesis testing is treated as a decision problem between two competing hypotheses:

- The null hypothesis H_0
- The alternative hypothesis H_1

Unlike Fisher, Neyman and Pearson defined two types of errors explicitly.

A Type I error occurs when the null hypothesis is rejected even though it is true. Its probability is defined as

$$\alpha = P(\text{Reject } H_0 \mid H_0) \tag{2.1}$$

A Type II error occurs when the null hypothesis is not rejected even though it is false. Its probability is denoted by β.

The significance level α must be fixed before observing the data. By fixing α in advance, the researcher controls the long run frequency of false positive decisions.

Neyman and Pearson also introduced the concept of statistical power, defined as

$$\text{Power} = 1 - \beta \tag{2.2}$$

Power represents the probability of correctly rejecting the null hypothesis when it is false. It depends on several factors, including the sample size, the true effect size, and the chosen significance level. Cohen (1988) later emphasized the importance of power analysis in study design [Coh88].

The Neyman-Pearson framework therefore treats hypothesis testing as a rule based decision procedure with controlled error probabilities. The focus is not on measuring evidence, but on minimizing incorrect decisions over repeated experiments.

2.3. Conceptual Differences Between Fisher and Neyman-Pearson

Although Fisher's and Neyman-Pearson's approaches are often presented together in modern statistics textbooks, they were originally developed

as distinct theoretical frameworks. Understanding their differences is important for interpreting the role of the significance level α.

Fisher's approach focused primarily on measuring the strength of evidence against the null hypothesis. In this framework, the p-value serves as a continuous indicator of how incompatible the data are with H_0. Fisher did not require the specification of a precise alternative hypothesis, nor did he formally define a second type of error. The emphasis was on interpreting the p-value within the context of a single study [Fis25].

In contrast, Neyman and Pearson (1933) developed a framework that treats hypothesis testing as a formal decision problem [NP33]. Their approach requires explicit specification of both the null and the alternative hypothesis. The significance level α is fixed in advance in order to control the long run probability of a Type I error. In addition, the probability of a Type II error, denoted by β, is explicitly defined, and statistical power is introduced as $1 - \beta$.

To summarize these conceptual differences, Table2.1 presents a direct comparison of the two approaches.

The fundamental distinction lies in interpretation. Fisher's method evaluates the strength of evidence in a single experiment, whereas the Neyman-Pearson framework evaluates the performance of a decision rule across repeated experiments.

	Fisher (1925)	Neyman–Pearson (1933)
Main focus	Measuring strength of evidence	Controlling decision errors
Role of p-value	Continuous measure of evidence	Compared to fixed α for decision
Alternative hypothesis	Not strictly required	Must be specified
Type I error (α)	Not central concept	Fixed in advance and controlled
Type II error (β)	Not formally defined	Explicitly defined
Interpretation	Single study inference	Long run error control

Table 2.1.: Comparison of Fisher and Neyman-Pearson approaches

2.4. The Institutionalization of $\alpha = 0.05$

Over time, the value $\alpha = 0.05$ became the default threshold in many scientific disciplines. The widespread use of this value was reinforced

through teaching materials, journal policies, and statistical software defaults.

Historical analysis suggests that the dominance of 0.05 is largely conventional rather than theoretically justified [CD82]. Nevertheless, it acquired the status of a standard rule. In many applied fields, results are routinely classified as either "statistically significant" or "not significant" based solely on whether the p-value falls below 0.05.

In recent years, this practice has been increasingly questioned. The American Statistical Association emphasized that scientific conclusions should not be based solely on whether a p-value crosses a fixed threshold [WL16]. Furthermore, proposals have been made to lower the default threshold for certain claims, for example to 0.005, in order to reduce false positive findings [Ben+18].

These discussions indicate that the choice of the significance level is not merely a technical detail, but a substantive methodological decision.

2.5. Implications for the Present Thesis

The historical development of hypothesis testing shows that the commonly used decision rule

$$\text{Reject } H_0 \text{ if } p \leq 0.05 \tag{2.3}$$

is the result of a combination of ideas from different theoretical frameworks and historical conventions.

Fisher emphasized evidence, whereas Neyman and Pearson emphasized error control. The modern routine use of a fixed significance level blends these perspectives without always acknowledging their conceptual differences.

This historical background provides the foundation for the present thesis. If the conventional significance level is historically contingent and theoretically flexible, then it is reasonable to explore whether a more adaptive and context sensitive approach to choosing α can improve statistical decision making.

The following chapter reviews how existing literature has approached this question and identifies the gap that motivates the present work.

Part III.

Literature on Significance Level Selection

3. Approaches in the Literature for Choosing the Significance Level

3.1. Conventional Fixed Significance Levels

In most empirical research, the significance level α is selected from a small set of conventional values, typically 0.10, 0.05, or 0.01. Among these, the value 0.05 has become the default standard across many disciplines. As discussed in Chapter 2, this convention can be traced back to Fisher's early work [Fis25], and its dominance was later reinforced through textbooks and journal practices [CD82].

The practical advantage of using a fixed threshold is simplicity. Researchers do not need to justify their choice of α for each study, and results can be compared easily across studies. However, this simplicity comes at a cost.

Mathematically, the conventional decision rule can be written as

$$\text{Reject } H_0 \text{ if } p \leq \alpha \tag{3.1}$$

When α is fixed at 0.05 for all studies, the probability of committing a Type I error is held constant at five percent in the long run. However, this does not guarantee balanced error rates in individual studies. In particular, the structure of the rejection rule in Equation 3.1 does not account for differences in sample size or expected effect size.

Statistical power as introduced in Chapter 1 can be expressed as

$$\text{Power} = P(\text{Reject } H_0 \mid H_0 \text{ is false}) = 1 - \beta \tag{3.2}$$

Equation 3.2 highlights the direct relationship between statistical power and the probability of a Type II error. Power is influenced by three main factors: the significance level α, the sample size N, and the true effect

size. [Coh88] emphasized that these elements should be considered jointly during study planning .

Although fixed thresholds provide comparability and simplicity, they do not adapt to differences between studies.

3.2. Power Based and Design Oriented Approaches

One line of research focuses on the relationship between the significance level and statistical power. In this view, the choice of α should not be isolated from study design decisions.

Consider a simple test statistic for a standardized effect size d. Under standard assumptions, the test statistic can be written as

$$Z = \frac{d\sqrt{N}}{\sigma} \tag{3.3}$$

Equation 3.3 shows that the magnitude of the test statistic increases with larger sample sizes and stronger effect sizes. The probability of rejecting the null hypothesis depends on how large Z is relative to the critical value determined by α.

If α is reduced, the critical value increases. This makes rejection more difficult and reduces the probability of a Type I error. However, as implied by Equation 3.2, reducing α may simultaneously decrease statistical power unless the sample size is increased accordingly.

From this perspective, some researchers argue that α should be chosen in conjunction with sample size planning. Although this approach improves study design, it typically still relies on selecting α from a small set of conventional values.

3.3. Decision Theoretic and Error Cost Approaches

Another class of approaches treats hypothesis testing explicitly as a decision problem. Within decision theoretic frameworks, the choice of the significance level α is based on the relative costs of different types of errors rather than on convention alone [Ber85].

In this setting, the objective is to select a decision rule that minimizes the expected loss associated with incorrect decisions. The expected loss depends on both the probabilities of error and the costs assigned to these errors.

Formally, following [Ber85], the expected loss function can be written as

$$L = C_1\alpha + C_2\beta, \tag{3.4}$$

where C_1 denotes the cost of a false positive decision and C_2 denotes the cost of a false negative decision. Since α and β denote the probabilities of Type I and Type II errors, respectively, Equation 3.4 expresses expected loss as a weighted combination of these two error probabilities.

Under this framework, the optimal significance level is the value of α that minimizes the expected loss. In principle, this provides a rational and context sensitive basis for selecting the test level.

Despite its theoretical appeal, practical implementation is often challenging. In many empirical research settings, the costs C_1 and C_2 cannot be quantified precisely, as they may depend on economic, ethical, or disciplinary considerations. For this reason, decision theoretic approaches are rarely applied in routine empirical research.

3.4. Recent Proposals for Stricter Thresholds

In response to concerns about replication failures, several authors have proposed lowering the conventional significance level. A prominent example is the recommendation to redefine statistical significance from 0.05 to 0.005 for claims of new discoveries [Ben+18].

Reducing α directly alters the rejection rule described in Equation 3.1. By lowering the threshold, the rejection region becomes smaller, thereby decreasing the probability of a Type I error.

However, as indicated by Equation 3.2, reducing α may also increase the probability of a Type II error unless adjustments are made to the study design. Thus, lowering the threshold does not eliminate the fundamental trade off between false positives and false negatives.

Importantly, even proposals such as $\alpha = 0.005$ still rely on selecting a fixed universal value.

3.5. Limitations of Existing Approaches

The literature demonstrates broad agreement that the universal use of $\alpha = 0.05$ is problematic. However, most proposed alternatives share two common limitations.

First, they typically rely on choosing from a small set of discrete values rather than treating α as a continuous parameter.

Second, they rarely provide a practical and easily implementable procedure that researchers can apply during study planning.

The structural trade offs between α and β, formalized in Equations 3.2 and 3.4, illustrate why a more systematic and adaptive approach may be desirable.

This gap motivates the development of the method proposed in this thesis. Rather than replacing the concept of statistical significance, the goal is to provide a systematic and data informed way of recommending a study specific significance level.

None of the reviewed approaches provide a practical, continuously valued, and data informed recommendation system that a researcher can use during study design without specialist expertise in decision theory or Bayesian statistics. The present thesis addresses this gap by combining empirical evidence from replication studies with a transparent rule based adjustment system, as described in Chapter 4.

Part IV.

System Development

4. Development of the Alpha Recommendation System

The main contribution of this thesis is the development of a data driven system for recommending study specific significance levels. Unlike conventional approaches that rely on fixed thresholds such as $\alpha = 0.05$, the proposed system combines empirical evidence from replication studies with machine learning and contextual statistical adjustments. This approach allows the recommended significance level to adapt to study characteristics and research preferences.

This chapter describes how the proposed Alpha Recommendation System is developed. The aim is to address the research gap identified in the previous chapters 3.5 by developing a practical and usable system.

This chapter explains step by step how the final alpha recommendation is generated. It starts with how the data are collected and prepared. Then it explains how the optimal alpha values are defined and how the prediction model is built. After that, it describes how user preferences and contextual adjustments are applied to produce the final recommended alpha. The recommended alpha is not a universal value but is tailored to the specific situation of the user's study. Together, these steps show how the system works from start to finish.

4.1. Overview of the System Design

The purpose of the system is to support researchers in selecting an appropriate significance level (α) for hypothesis testing. Instead of relying on a fixed conventional value such as $\alpha = 0.05$, the system provides a recommended α value based on study characteristics and research context. The recommendation is intended to guide decision making during the study design stage and does not replace statistical judgment.

The system is implemented as a software application in Python and includes a machine learning component. It operates in two stages: a data driven regression model first estimates a baseline significance level (α_{base}) from sample size and effect size, and a context adjustment step

then refines this value based on researcher preferences. The two stages are described in detail in Sections 4.1.1 and 4.1.2 respectively, and the full system architecture is illustrated in Figure 4.1.

The overall structure of the Alpha Recommendation System is illustrated in Figure 4.1.

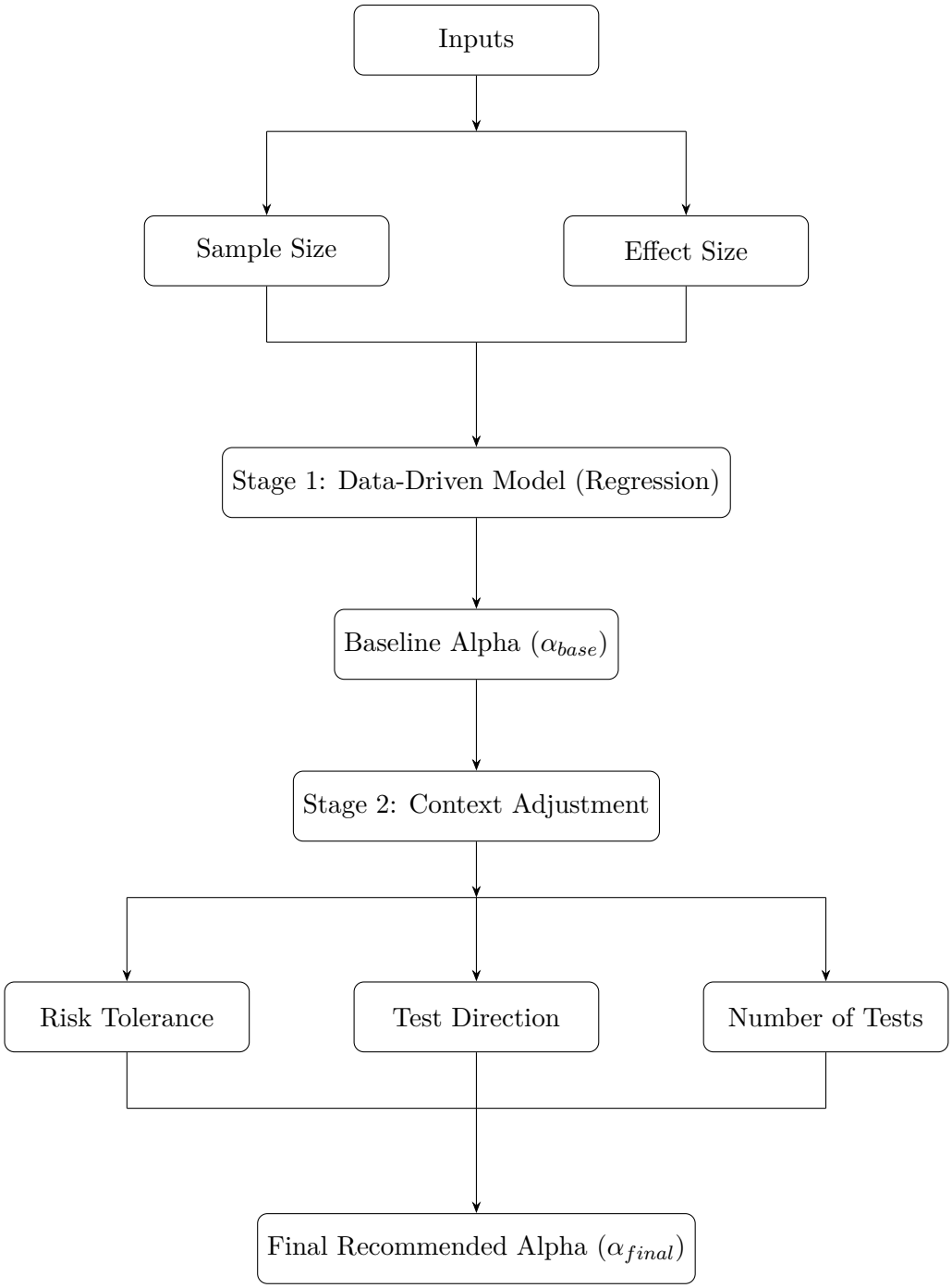

Figure 4.1.: Flow diagram of the Alpha Recommendation System.

4.1.1. Data Driven Alpha Recommendation

The first stage of the Alpha Recommendation System predicts a baseline value for the significance level (α) using information from past replication studies. This stage relies only on information that is available before performing the hypothesis test for the new study.

Two inputs are used: sample size and effect size. These variables are chosen because they are specified during study planning and influence statistical power and error rates[PF21]. No p-values or results from the new study are used at this stage.

Replication data are used to examine how different combinations of sample size and effect size are associated with appropriate α values in previous studies. Based on these patterns, the system learns a general relationship between these inputs and corresponding alpha values.

When a researcher provides the sample size and expected effect size for a new study, the system applies this learned relationship to produce a baseline α value. This value reflects patterns observed in similar past studies and serves as a starting point rather than a final decision.

The internal data preparation and model training procedure used to obtain this α_{base} value are described in a later sections 4.2 and 4.3.

4.1.2. Context Based Adjustment of Alpha

The α_{base} value produced in the first stage does not include all aspects of a study. Accurate decisions depend on the research context and the researcher's preferences in addition to the patterns observed in replication data which provide only a numerical baseline and do not account for the specific context or preferences of the researcher. For this reason, the Alpha Recommendation System includes a second stage that adjusts the α_{base} value using user defined inputs.

In this stage, the user provides information about the study context. Three factors are considered: risk tolerance, test directionality (one tailed or two tailed), and the number of hypotheses being tested. These factors were selected because they are commonly used in classical hypothesis testing and have a clear connection to the choice of the significance level.

The system applies simple and predefined rules to modify the α_{base} value. Stricter risk preferences lead to smaller α values, while more lenient preferences allow larger values[Bak16].

The first adjustment modifies the baseline alpha according to the selected

level of risk tolerance

$$\alpha' = \alpha_{\text{base}} \cdot k_1 \tag{4.1}$$

where α' denotes the risk-adjusted baseline alpha and k_1 is the risk tolerance scaling factor. One tailed tests and multiple hypothesis testing are handled using standard statistical adjustments.

The second adjustment accounts for the structural difference between one tailed and two tailed tests[Hal23]:

$$\alpha'' = \begin{cases} 2\alpha' & \text{if one-tailed} \\ \alpha' & \text{if two-tailed} \end{cases} \tag{4.2}$$

where α'' denotes the directionality adjusted alpha value. This adjustment reflects the allocation of the rejection region in hypothesis testing.

Finally, when multiple hypotheses are tested simultaneously, a Bonferroni-style correction is applied[HFC19]:

$$\alpha_{\text{final}} = \frac{\alpha''}{m} \tag{4.3}$$

where m denotes the number of hypothesis tests conducted.

To ensure numerical stability and interpretability, the final alpha value is restricted to the interval:

This prevents unrealistically small or excessively large significance levels.

$$0.001 \leq \alpha_{\text{final}} \leq 0.15 \tag{4.4}$$

The rules are transparent, so the effect of each input on the final α value is easy to understand.

Although only three contextual factors are included, the system remains broadly usable. The α_{base} value, which is derived from replication data, can serve as a default and data informed reference for selecting the significance level. The contextual adjustments are optional refinements that allow the recommendation to better reflect specific study preferences. Even if additional contextual factors are not included, the system still provides a more flexible and evidence based alternative to using a fixed value such as $\alpha = 0.05$. The overall adjustment process can be summarized mathematically as follows:

$$\alpha_{\text{final}} = \alpha_{\text{baseline}} \cdot k_1 \cdot k_2 \cdot k_3 \tag{4.5}$$

where k_1 represents the risk preference factor, k_2 represents the test direction factor, and k_3 represents the multiple testing correction factor.

After applying the adjustments, the system produces a final recommended α value. This value is selected before conducting the hypothesis test and does not depend on observed data or p-values. It does not replace statistical judgment, but instead provides a structured and context aware reference for choosing the significance level.

The implementation details used to perform these adjustments are described in the following section.

4.2. Dataset Construction and Optimal Alpha Definition

This section explains how the dataset used in the Alpha Recommendation System is constructed. It describes the data sources, their reliability, the procedure used to define optimal alpha values, the methods used to recover missing sample sizes and effect sizes, and the structure of the final dataset.

The dataset used in the Alpha Recommendation System is constructed by combining results from several large scale replication projects available through the Open Science Framework (OSF). These projects report findings from original studies together with the results of independent replication attempts across multiple research fields[FD17]. Each row in the final dataset represents a single hypothesis test.

In many original studies, the significance level (α) is not explicitly reported. This is because most studies follow the common convention of using $\alpha = 0.05$[Kim15]. When a different significance level is used, it is usually stated clearly in the study. Therefore, when no α value is reported, it is assumed that the original study used $\alpha = 0.05$. This assumption reflects common reporting practice in empirical research.

For studies in which the replication confirms the original finding, the significance level used in the original study is treated as appropriate. In these cases, the original α value (either reported or assumed to be 0.05) is recorded as the optimal alpha, since the original decision is supported by the replication outcome.

For studies in which the replication does not confirm the original finding, the original significance level is considered too lenient. In such cases, an alternative α value is calculated using the original study's reported p-value. The purpose is to determine which significance level would have led to a correct decision when compared with the replication result. The optimal alpha is defined as the largest significance level that is smaller than the original p-value. This ensures that the original result would not

have been declared statistically significant under the adjusted threshold. The calculations are explained in detail in Section 4.2.6.

This procedure generates a study specific optimal alpha value for each hypothesis test in the dataset. The calculation relies only on information reported in the original and replication studies, including p-values, test statistics, sample size, and effect size. When necessary, standard statistical relationships are used to ensure consistency across different study designs.

After applying this procedure to all replication projects, the result is a single combined dataset in which each row represents one hypothesis test and includes a clearly defined optimal alpha value. This optimal alpha serves as the target variable for the data driven component of the Alpha Recommendation System. The detailed steps of dataset construction and the full procedure for defining the optimal alpha are explained in the following sections.

4.2.1. Data Acquisition

The dataset used in this study is constructed by combining results from five large scale replication projects available on the Open Science Framework (OSF). These projects report findings from original studies together with the results of independent replication attempts across multiple research domains[FD17]. Each row in the final dataset used in this project represents a single hypothesis test.

Table 4.1.: Overview of replication projects included in the dataset

Replication Project	**Research Field**	**Main Variables Extracted**
Reproducibility Project: Cancer Biology [ED21]	Biomedical	Original p-values, standardized effect sizes, sample sizes, and replication p-values.
Challenging the N-Myopia[Li21]	Psychology	Original p-values, effect sizes, sample sizes, and replication success indicators.
Applying Z-Curve to Large-Scale Replication Studies [Sot22]	Psychology	Effect sizes and sample sizes derived from reported test statistics together with replication outcomes.
Pre-hacking Project[Cha25]	Psychology	Original p-values, recovered effect sizes, sample sizes, and replication results.
Life Outcomes of Personality Replication (LOOPR)[Sot17]	Personality / Outcomes	Preregistered effect sizes, sample sizes, and replication outcomes.

4.2.2. Data Quantity

After cleaning and compilation, the final dataset contains 664 individual hypothesis tests drawn from five replication projects. The contributions

from each project are as follows:

- Cancer Biology Replication Project: 135 tests
- N-Myopia Replication Project: 310 tests
- Z-Curve Replication Studies: 136 tests
- LOOPR Replication Project: 75 tests
- Pre-hacking Replication Project: 8 tests

Combining multiple replication projects increases diversity in study design and reduces reliance on a single research domain. Sample sizes in the dataset range from very small studies to large scale experiments with sample sizes exceeding 200,000. Effect sizes also cover a wide range, including both small and large effects.

The size of the dataset is sufficient for the purpose of this study because the goal is not high dimensional prediction, but learning stable relationships between a small number of predictors—primarily sample size and effect size—and corresponding optimal alpha values. The dataset provides adequate variation across study contexts while remaining manageable and interpretable.

4.2.3. Data Quality and Reliability

The Open Science Framework is widely used for sharing replication data because it promotes transparency and open research practices. Replication studies hosted on OSF are typically conducted by independent research teams, often follow preregistered analysis plans, and provide open access to data and study materials, replication projects hosted on OSF are conducted by established research teams and are published in peer reviewed journals, which ensures that the results have been scientifically reviewed and validated[FD17]. These characteristics increase the credibility of the replication outcomes and make OSF projects a reliable source for evaluating statistical decision rules such as hypothesis testing.

The selected projects cover different research fields, including psychology, behavioral science, personality research, and biomedical science. This cross disciplinary selection ensures diversity in study designs and increases the generalizability of the constructed dataset. Each project provides hypothesis level information including p-values, effect sizes, sample sizes, and replication outcomes.

4.2.4. Data Relevance

Replication data are directly relevant because they allow evaluation of whether original statistical decisions were supported by subsequent evidence[FD17]. By linking original p-values, sample sizes, effect sizes, and replication outcomes, the dataset provides a structured basis for analyzing how different significance levels relate to reproducibility..

4.2.5. Handling Missing Values and Anomalies

Not all studies reported complete information. Some studies did not explicitly report sample size or effect size but provided test statistics such as t, F, or χ^2 values. Rather than discarding these studies, standard statistical relationships were used to recover missing quantities when sufficient information was available. The methods are explained in 4.2.7

Studies that lacked enough information to reliably recover sample size or effect size were excluded. No aggressive outlier removal was applied. Only clearly implausible or incomplete records were removed to maintain consistency while preserving as much valid replication information as possible.

4.2.6. Definition and Calculation of Optimal Alpha

In some original studies, the significance level (α) used for hypothesis testing is not explicitly reported. This is because most studies follow the conventional choice of $\alpha = 0.05$ [Kwa23]. When an alternative significance level is used, authors typically report it explicitly. Therefore, when no α value is reported, it is assumed that the original study used $\alpha = 0.05$. For studies in which the replication successfully confirms the original finding, the original significance level is treated as appropriate. In these cases, the original α value (reported or assumed) is recorded as the optimal alpha, since the original decision is supported by the replication outcome.

For studies in which the replication does not confirm the original finding, the original significance level is considered too lenient.Concerns about overly permissive significance thresholds and their impact on replication have been widely discussed in the statistical literature [Ben+18; WL16]. In such cases, an alternative significance level is derived using the original study's reported p-value p_{orig}. The objective is to determine a study specific significance level that would have resulted in a non significant outcome in the original study while remaining as close as possible to the reported p-value.

Concretely, the optimal alpha is defined as

$$\alpha_{\text{opt}} = \max\{\alpha \in \mathcal{A} : \alpha < p_{\text{orig}}\}, \tag{4.6}$$

where

$$\mathcal{A} = \{0.001, 0.005, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.09, 0.10\}$$

This candidate set was derived empirically from the replication dataset. The optimal alpha values observed across all 664 studies fell within this discrete range, making these values the natural choice for the candidate set.

This Formula4.2.6 was derived specifically for this system to operationalise the concept of a study-specific threshold. While the general idea of adapting significance levels to study outcomes is discussed in the literature [Ben+18][WL16], the specific formulation presented here was developed as part of this work.

is a predefined set of candidate significance levels. This rule ensures that the original result would not have been declared statistically significant under α_{opt}, while the chosen value remains interpretable and comparable across studies.

For example, if a study has $p = 0.047$ and the replication did not find the effect, using $\alpha = 0.05$ would incorrectly declare the result significant. Using $\alpha = 0.04$ would correctly declare it non significant. In this case, $\alpha = 0.04$ is the optimal alpha for that study.

This procedure produces a study specific optimal alpha value for every hypothesis test in the dataset. The optimal alpha serves as the target variable for the data driven component of the Alpha Recommendation System.

It is important to note that replication outcomes alone cannot determine a theoretically optimal significance level. The procedure used here should therefore be interpreted as a pragmatic approximation. Similar discussions about adapting significance thresholds in response to replication evidence can be found in the statistical literature on reproducibility and significance testing [Ben+18; WL16; Ope15].

4.2.7. Derivation of Sample Size and Effect Size

To ensure consistency across studies, each hypothesis test must be associated with a sample size and an effect size. When these values were not reported directly, they were recovered using standard statistical formulas.

For t-tests with reported degrees of freedom, effect size r was computed as [Bec00]:

$$r = \sqrt{\frac{t^2}{t^2 + df}} \tag{4.7}$$

If sample size was not explicitly reported but degrees of freedom were available, sample size was recovered as[Coh88]:

$$N = df + 2 \tag{4.8}$$

For χ^2 tests with one degree of freedom, effect size was computed using [Kim17]:

$$\phi = \sqrt{\frac{\chi^2}{N}} \tag{4.9}$$

For F-tests with one numerator degree of freedom, effect size was computed as [Ric11]:

$$\eta^2 = \frac{F \cdot df_1}{F \cdot df_1 + df_2}, \qquad r = \sqrt{\eta^2} \tag{4.10}$$

Only studies for which both sample size and effect size could be reliably obtained were retained. A complete list of formulas is provided in Appendix A.

4.2.8. Final Dataset Structure

After combining replication projects, computing optimal alpha values, and recovering missing sample sizes and effect sizes, the final dataset consists of standardized rows. Each row represents one hypothesis test.

Table 4.2 shows the first five rows of the final dataset to illustrate its structure.

The final dataset is used as input for the machine learning component of the Alpha Recommendation System. Sample size and effect size are used as predictors, and optimal alpha is used as the target variable.

It should be noted that larger sample sizes do not always correspond to stricter optimal α values in the dataset. In cases where the effect size is relatively large and the original p-value was substantially above a conventional threshold, the calculated optimal α may exceed 0.05 even

Table 4.2.: First Five Rows of the Final Dataset

Sample Size	Effect Size	Replication Success	Best Alpha
30	0.18	0	0.031
120	0.25	1	0.050
85	0.14	0	0.042
300	0.36	1	0.050
1000	0.62	0	0.090

when the replication failed. This reflects the specific p-value reported in the original study rather than a general recommendation for leniency.

4.3. ML Model for Predicting α_{base}

This section explains how the constructed dataset is processed specifically for model training, how the input features and target variable are selected, and how the data driven model is implemented to estimate a α_{base}.

While the previous section described how the dataset was built and how optimal alpha values were defined4.2, this section focuses on preparing the data for machine learning and implementing the predictive model.

4.3.1. Data Preparation for Model Training

After the dataset is constructed and the optimal alpha values are defined, it is prepared for use in the machine learning model. The purpose of this step is to ensure that the data are suitable for training while preserving the original information as much as possible.

Only rows that contain all required variables are retained. Each observation must include:

- Valid sample size,
- Valid effect size,
- Defined optimal alpha value.

Rows are removed only when essential information is missing and cannot be recovered. For example, if sample size or effect size cannot be derived from the available statistics, that observation is excluded.

Clearly invalid values are also removed. Observations with zero or negative sample sizes, undefined effect sizes, or inconsistent values are excluded to

prevent errors during model training.

No additional filtering or transformation is applied beyond these basic checks. Sample sizes are not trimmed, and effect sizes are not rescaled unless required during calculation. This approach preserves the natural variation in the replication data and allows the model to learn from real differences between studies.

After these steps, the dataset contains complete and valid observations suitable for training the predictive model.

4.3.2. Feature Selection and Target Variable

The goal of the model is to recommend a significance level before conducting a hypothesis test. Therefore, only information known at the study design stage is used as input.

Two input features are selected:

- Sample size
- Effect size

Sample size is included because it directly affects statistical power and the balance between Type I and Type II errors. Larger studies can often support stricter significance levels, while smaller studies may require more flexible thresholds [ML22].

Effect size is included because it reflects the strength of the expected relationship. Studies targeting small effects face different decision challenges than those targeting larger effects [Mud+12].

The model predicts the optimal alpha value as a continuous variable. This optimal alpha represents the significance level that would have produced a decision consistent with the replication outcome.

Contextual factors such as risk tolerance, test directionality, and multiple testing are not included in this stage because they are handled in stage two of the application and explained step by step in following sections 4.4.

4.3.3. Conceptual Idea of the Data Driven Model

The purpose of the data driven model is to estimate a baseline alpha value based on patterns observed in replication studies. The model does not make final decisions about hypothesis testing. Instead, it provides a reference value based on empirical evidence.

Each training example (Each row in the dataset) consists of:

- Sample size (input)
- Effect size (input)
- optimal alpha value (target).

The model learns how changes in sample size and effect size are associated with changes in suitable significance levels. If studies with certain characteristics required stricter or more lenient alpha values to align with replication outcomes, the model captures this pattern.

When new study inputs are provided, the model produces a α_{base} value based on pattern learned from similar past cases.

4.3.4. Choice of Algorithm

The predictive model is implemented in Python using the scikit-learn library. Scikit-learn is a widely used open source machine learning library that provides tools for data preprocessing, model training, evaluation, and prediction. It allows the implementation of various machine learning algorithms in a structured and efficient way [Sci23].

Since the objective is to predict a continuous optimal alpha value, the problem is formulated as a regression task. Several machine learning algorithms were tested for this purpose, including Decision Tree, Random Forest, and XGBoost regression models. Their predictive performance was compared using standard evaluation metrics 4.2, 4.3,4.4.

To evaluate predictive performance, the dataset was divided into training and testing subsets. Eighty percent of the observations were used to train the models, while the remaining twenty percent were reserved for testing. Model performance was evaluated on the test set using the Mean Absolute Error (MAE) metric.

The Mean Absolute Error (MAE) is defined as

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.11}$$

where y_i represents the true value and $\hat{y}_i$ represents the predicted value[WM05].

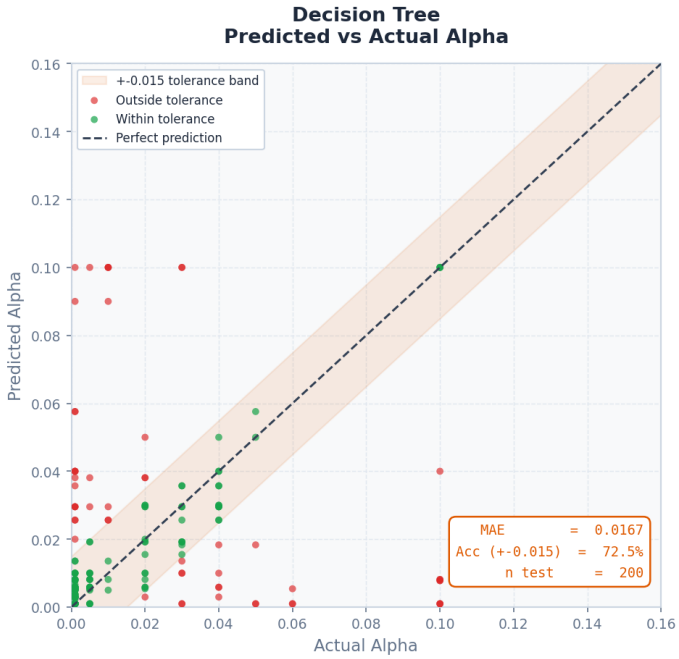

Figure 4.2.: Decision Tree Regressor: Predicted/Actual Alpha Values

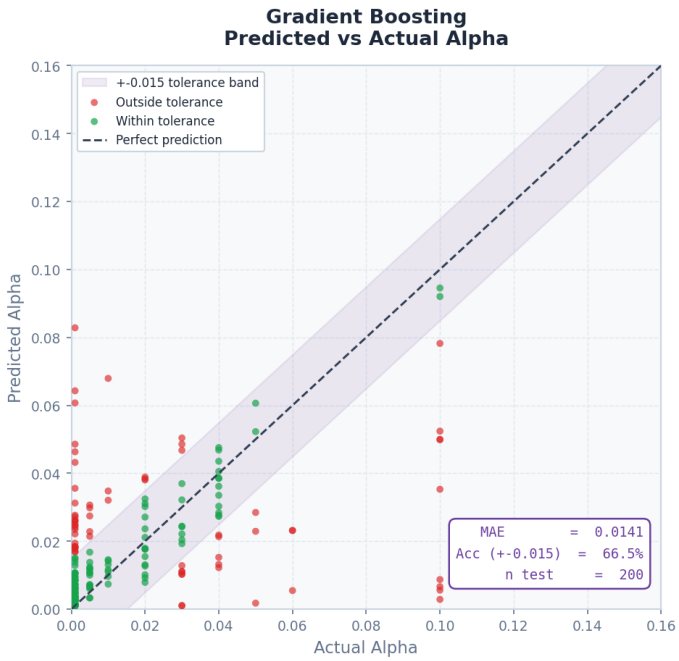

Figure 4.3.: Gradient Boosting Regressor: Predicted/Actual Alpha Values

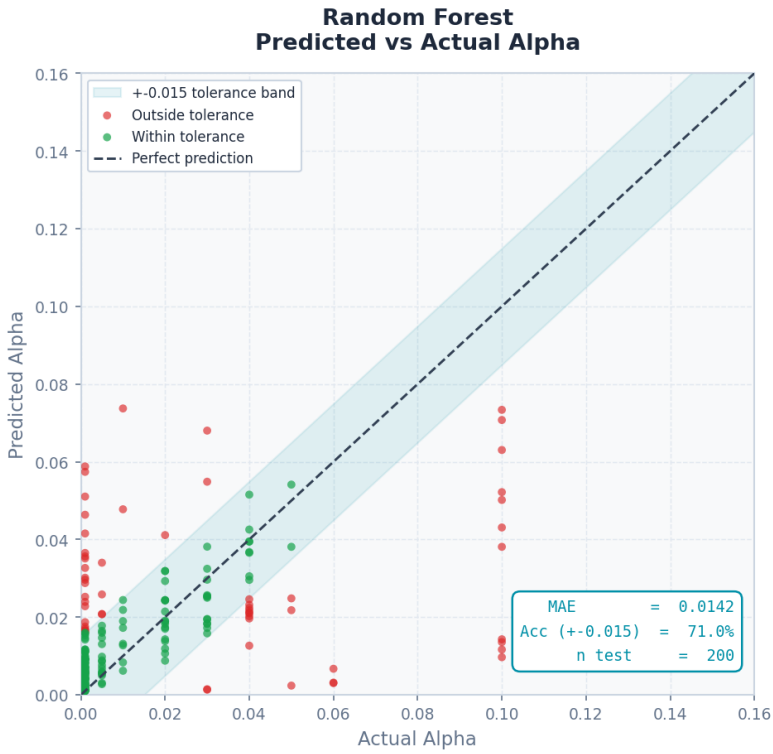

Figure 4.4.: Random Forest Regressor: Predicted/Actual Alpha Values

Among the tested models, the Random Forest regression algorithm achieved the lowest Mean Absolute Error (MAE) of 0.0142 on the test set ($n = 208$), with 71.0% of predictions falling within a ± 0.015 tolerance band. This outperformed the Decision Tree (MAE = 0.0167, 72.5% within tolerance) and Gradient Boosting (MAE = 0.0141, 66.5% within tolerance) regressors.

Random Forest builds multiple decision trees and combines their predictions by averaging and better explained in Figure4.5[Bre01]. This approach better

- captures non linear relationships,
- reduces sensitivity to noise,
- and does not require strong distributional assumptions.[RRS17]

The objective is not to develop a new algorithm, but to use a reliable and widely accepted method to learn patterns from replication data.

The predictive performance of the models was evaluated using the regression metric Mean Absolute Error (MAE). MAE measures the average absolute difference between the predicted and actual values. This metric was chosen because it is easy to understand and expresses the prediction

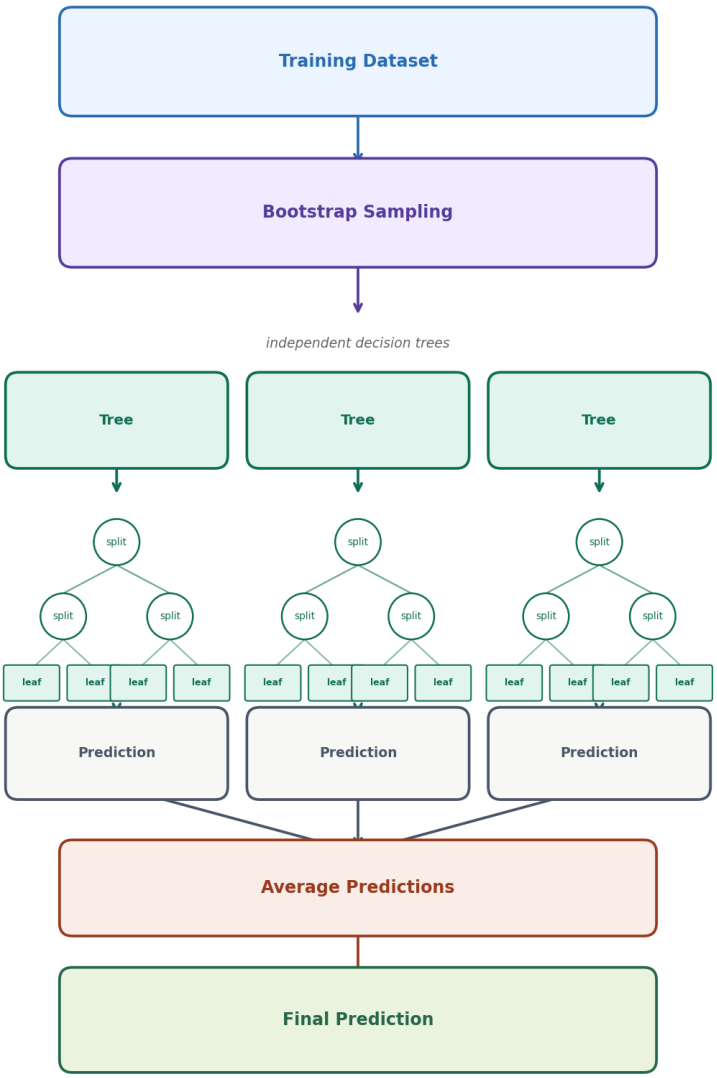

Figure 4.5.: Structure of the Random Forest regression model used for predicting the baseline significance level.(based on [Ha23])

error in the same units as the target variable. In addition, MAE is less sensitive to large errors than metrics such as Mean Squared Error (MSE), since it does not square the differences between predicted and actual values [WM05].

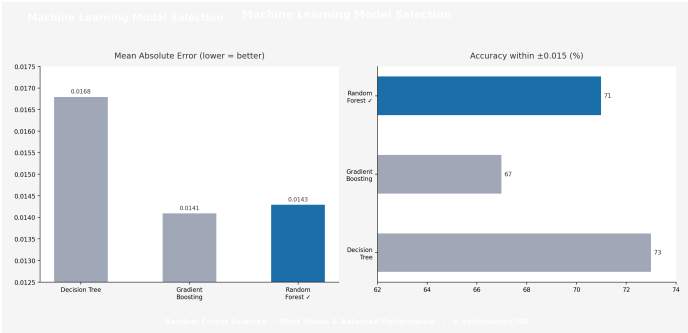

Figure 4.6.: Performance comparison of ML models

Among the tested models, the Random Forest regression algorithm achieved the lowest MAE4.4, indicating better predictive performance compared to the alternative models.

4.3.5. Model Implementation in Python

The implementation is written in Python using standard data analysis and machine learning libraries, including **pandas for data handling, NumPy** for numerical operations, and **scikit-learn** for model training and evaluation. Each step of the implementation, including data preparation, model training, prediction, and evaluation, is described in detail in the following sections.

Loading the Data

```
import pandas as pd

data = pd.read_excel("master_alpha_dataset_updated.xlsx")

X = data[["sample_size", "effect_size"]]
y = data["best_alpha"]
```

The dataset is loaded from an Excel file using the pandas library. The variables **sample size** and **effect size** are selected as input features (X), and **best alpha** is selected as the target variable (y). This separation allows the model to learn the relationship between study characteristics and the optimal alpha value.

Training the Model

```
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor(n_estimators=200, random_state
    =42)
model.fit(X, y)
```

Training the Model

The Random Forest regression model was implemented using the `scikit-learn` library in Python. The dataset was divided into training and test sets using an 80/20 split. This proportion is commonly used in machine learning because it balances the need for sufficient training data with the need for an independent evaluation dataset.

The number of trees in the forest was set to 200. Increasing the number of trees generally improves the stability of ensemble models, but after a certain point the performance gains become negligible. Preliminary experiments indicated that performance stabilised beyond approximately 150 trees, so 200 estimators were selected.

The random state was fixed at 42 to ensure reproducibility of the results. This guarantees that the same training test split and model initialization can be reproduced in subsequent runs.

Predicting α_{base}

```
predicted_alpha = model.predict([[150, 0.30]])
```

After training, the model can predict a α_{base} for new inputs. When a sample size of 150 and an expected effect size of 0.30 are provided, the model returns a predicted alpha value based on patterns learned from the replication data.

4.4. Context Based Adjustment of the α_{base}

This section describes the second stage of the Alpha Recommendation System. In this stage, the α_{base} value produced by the data driven model is adjusted using user defined preferences with standard statistical rules. The purpose of this step is to update α_{base} to become context aware and to suggest user an alpha to their preference.

4.4.1. Motivation for Context Based Adjustment

The α_{base} predicted by the model reflects patterns observed in replication studies. However, no single alpha value can be appropriate for all research situations. Even when sample size and effect size are similar, studies may differ in risk tolerance, testing strategy, or research goals .

For this reason, the system includes a separate adjustment step. This ensures that contextual preferences are applied explicitly after the data driven prediction, rather than influencing the model itself. This separation maintains transparency and interpretability. The overall adjustment formula, introduced in Section 4.1, combines these three factors as follows (see Equation 4.12):

$$\alpha_{\text{final}} = \alpha_{\text{base}} \cdot k_1 \cdot k_2 \cdot k_3 \tag{4.12}$$

where k_1 represents the risk preference factor, k_2 represents the test direction factor, and k_3 represents the multiple testing correction factor.

4.4.2. Adjustment Factors

Three contextual factors are included:

1. Risk tolerance
2. Test directionality (one tailed or two tailed)
3. Number of hypotheses tested

These factors were selected because they are commonly considered in classical hypothesis testing and have a direct and well defined relationship with the significance level[Bak16][Hal23][HFC19].

Although other contextual factors may exist, the system focuses on these widely used adjustments to maintain simplicity and clarity. The α_{base} remains usable as a general reference even without additional adjustments.

4.4.3. Risk Tolerance

Risk tolerance reflects how strict or lenient a researcher wants to be when controlling false positives (Type I errors). A stricter approach uses a smaller alpha value, while a more exploratory approach may allow a larger alpha value[Bak16].

To make this preference easy to use, the system provides five qualitative levels:

- very strict
- strict
- neutral
- lenient
- very lenient

Five levels are chosen to give a balanced range of options. They allow the researcher to move gradually from very cautious to more flexible decisions, without making the system overly complex.

$$\alpha' = \alpha_{\text{base}} \cdot k_1 \tag{4.13}$$

Each level is converted into a numerical adjustment by multiplying the α_{base} with a fixed scaling factor. This converts a qualitative choice (for example, "strict") into a clear mathematical change.

The adjustment is implemented as follows:

```
def adjust_for_risk(alpha, risk_level):
if risk_level == "very_strict":
return alpha * 0.5
elif risk_level == "strict":
return alpha * 0.75
elif risk_level == "lenient":
return alpha * 1.25
elif risk_level == "very_lenient":
return alpha * 1.5
return alpha
```

If *neutral* is selected, the alpha value remains unchanged.

Multiplication is used because alpha is a probability value. Proportional changes keep the adjustment consistent across different baseline values. For example, if the α_{base} is 0.04:

- Very strict ($\times 0.5$) $\rightarrow$ 0.02
- Strict ($\times 0.75$) $\rightarrow$ 0.03
- Neutral $\rightarrow$ 0.04
- Lenient ($\times 1.25$) $\rightarrow$ 0.05
- Very lenient ($\times 1.5$) $\rightarrow$ 0.06

The scaling values used in this thesis (0.5 for "very strict" risk tolerance, 0.75 for "strict", 1.25 for "lenient", and 1.5 for "very lenient") are heuristic but follow two design principles. First, they are symmetric around 1, so that stricter and more lenient choices adjust the α_{base} by comparable

relative amounts. Second, the factors are moderate in magnitude, ensuring that the resulting α values remain within a practically interpretable range defined in Equation 4.4. These values were selected as a practical starting point consistent with proportional reasoning used in decision theoretic frameworks [Ber85]. They are not empirically derived and represent a deliberate design simplification to maintain interpretability. Future work could calibrate these factors empirically across a larger collection of replication studies.

4.4.4. Test Directionality

After adjusting the α_{base} for risk tolerance, the system next considers the direction of the hypothesis test.

Before conducting a hypothesis test, the researcher must decide whether the test is one tailed or two tailed. In a two tailed test, the total significance level is divided between both sides of the distribution. In a one tailed test, the entire significance level is placed in only one direction.[Hal23]

$$\alpha'' = \begin{cases} 2\alpha' & \text{if one-tailed} \\ \alpha' & \text{if two-tailed} \end{cases} \tag{4.14}$$

To reflect this structural difference, the system adjusts the alpha value when a one tailed test is selected. The adjustment is implemented in Python as follows:

```
def adjust_for_tail(alpha, tail_type):
if tail_type == "one_tailed":
return alpha * 2
return alpha
```

If a one tailed test is chosen, the alpha value is multiplied by two. If a two tailed test is selected, no adjustment is made.

If no direction is specified, the system assumes a two tailed test and leaves the alpha unchanged.

4.4.5. Multiple Testing

When multiple hypotheses are tested within the same study, the probability of at least one false positive increases. To account for this, the system applies a Bonferroni-style correction[HFC19].

$$\alpha_{\text{final}} = \frac{\alpha''}{m} \tag{4.15}$$

```
def adjust_for_multiple_tests(alpha, num_tests):
if num_tests > 1:
return alpha / num_tests
return alpha
```

Dividing the alpha value by the number of tests makes the decision rule more conservative as the number of comparisons increases.

4.4.6. Combining the Adjustments

The final alpha(α_{final}) value is obtained by applying the adjustment rules sequentially to the α_{base}:

```
alpha_final = adjust_for_risk(alpha_base, risk_level)
alpha_final = adjust_for_tail(alpha_final, tail_type)
alpha_final = adjust_for_multiple_tests(alpha_final,
    num_tests)
```

Each step corresponds to a clear design choice made by the researcher.

To prevent unrealistic values, the α_{base} is restricted to a predefined range:

```
alpha_final = max(0.001, min(alpha_final, 0.15))
```

$$0.001 \leq \alpha_{\text{final}} \leq 0.15 \tag{4.16}$$

This ensures that the recommended alpha remains within reasonable and interpretable limits.

4.4.7. Understanding the Final Alpha Recommendation

The α_{base} value should be understood as a recommendation rather than a strict rule. It combines information learned from replication data with preferences defined by the researcher. The predictive model first estimates a α_{base} based on patterns observed in previous studies.

In the second step, the researcher can adjust this value according to the needs of the study, such as the desired level of strictness or risk tolerance. This makes the recommendation flexible and easier to apply in different research situations. The goal is not to define a theoretically optimal alpha level, but to provide a practical and understandable guideline.

4.4.8. Role of the Adjustment Step in the System

The context based adjustment represents the second and final stage of the Alpha Recommendation System. In the first stage, the machine learning

model predicts a α_{base} using empirical data. In the second stage, the researcher can modify this value to reflect study specific considerations.

By separating these two stages, the system remains transparent and easy to interpret. Researchers can clearly see which part of the recommendation comes from the data and which part results from their own preferences. This design ensures that the system provides guidance while still allowing researchers to retain full control over their final decision.

Part V.

Deployment

5. Deployment

Before describing the technical implementation of the deployment, it is useful to illustrate how the deployed system operates from a user perspective. Figure 5.1 illustrates the overall workflow of the deployed Alpha Recommendation System, showing how the trained model, contextual adjustments, and the user interface interact to produce a final α recommendation.

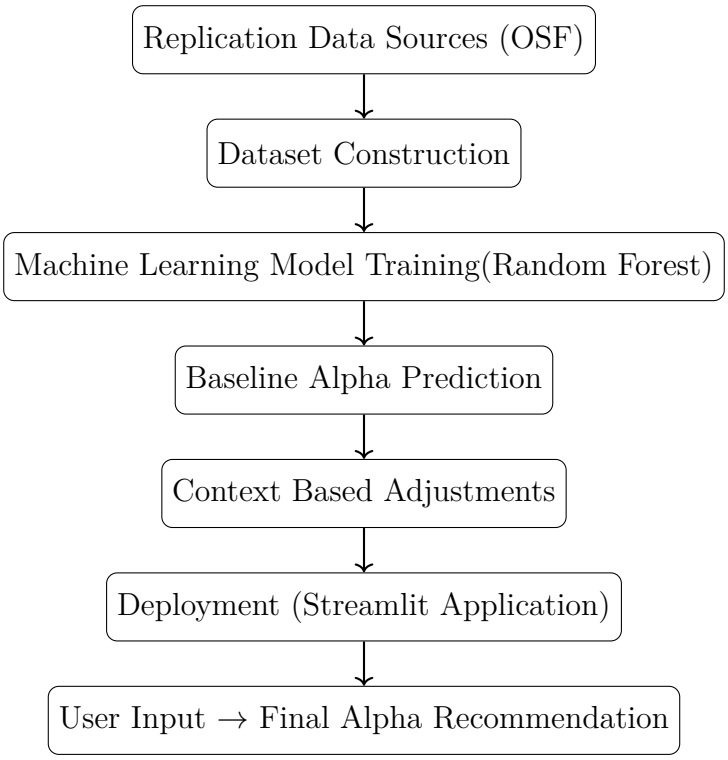

Figure 5.1.: Machine Learning System Architecture

5.1. Model Deployment

The Alpha Recommendation System is deployed as an interactive application so that researchers can easily obtain an alpha recommendation

without writing code. The deployed system allows users to enter study parameters and immediately receive a recommended significance level.

Figure 5.1 illustrates the overall workflow of the deployed system. The trained model developed in Chapter 4 is integrated into a Streamlit application. The following sections describe how users interact with the system to obtain an alpha recommendation.

5.1.1. Purpose of Deployment

The purpose of deployment is to make the developed model usable for researchers during the study planning stage. Instead of running Python scripts manually, users interact with the system through a simple interface for entering study parameters and receive a recommended significance level.

5.1.2. Choice of Deployment Tool

The system is deployed using Streamlit, a Python framework for building interactive data applications. Streamlit allows a complete interface to be created directly from Python code without requiring additional web development technologies such as HTML or JavaScript[Str24].

This makes the system easy to run locally and accessible to users with limited programming experience.

5.1.3. Loading the Trained Model

After the machine learning model is trained during system development, it is saved as a serialized `.pkl` file using the Python *pickle* library. This file stores the trained model and its learned parameters.

During deployment, the Streamlit application loads the stored model when the application starts. This avoids retraining the model every time the system is used and allows predictions to be generated immediately.

The trained model is loaded using the following code:

```
import pickle

with open("final_alpha_recommendation_system.pkl", "rb") as
    file:
model = pickle.load(file)
```

Once loaded, the model is ready to generate predictions based on user inputs.

When the Streamlit application starts, this code loads the serialized model into memory. Once loaded, the model becomes available to the application and can immediately generate predictions based on user inputs. This approach improves efficiency because the computationally expensive training step does not need to be repeated each time the application is executed.

5.2. User Interaction

After the model is loaded, the application allows users to interact with the system through the Streamlit interface. Users provide the study parameters required for generating an alpha recommendation. The system then processes these inputs and returns the final result.

The interaction between the stored model, the Streamlit interface, and the prediction pipeline is illustrated in Figure 5.3

5.2.1. User Input Interface

The application provides a simple interface where users can enter the required study parameters. These include:

- sample size
- expected effect size
- risk tolerance
- test type (one or two tailed)
- number of hypothesis tests

Figure 5.2 shows the interface before the recommendation is generated.

The following code defines the input components of the Streamlit application:

```
import streamlit as st

sample_size = st.number_input("Sample size", min_value=1)
effect_size = st.number_input("Effect size", min_value=0.0)
risk_level = st.selectbox("Risk tolerance",
["very_strict", "strict", "neutral",
"lenient", "very_lenient"])
tail_type = st.selectbox("Test type",
["two_tailed", "one_tailed"])
num_tests = st.number_input("Number of hypothesis tests",
min_value=1)
```

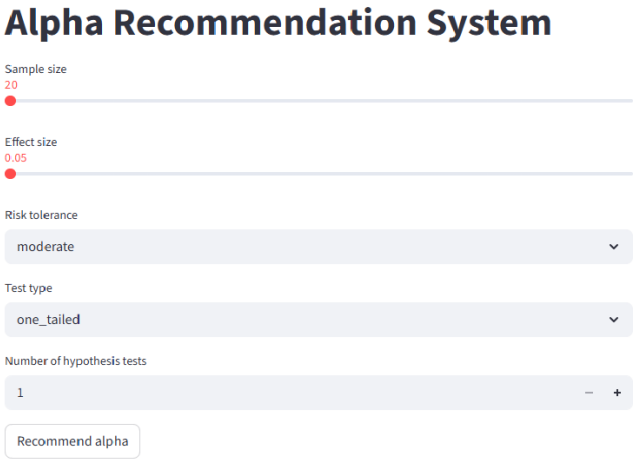

Figure 5.2.: Interface of the system before entering the values

These inputs are passed to the prediction pipeline to generate the recommendation.

5.2.2. Generating the Alpha Recommendation

Once the user enters the required parameters, the application generates the alpha recommendation in two steps.

- First, the trained machine learning model predicts a α_{base} value using the input sample size and expected effect size.
- Second, the system applies contextual adjustments based on the selected risk tolerance, test type, and number of hypothesis tests.

The implementation is shown below:

```
alpha_base = model.predict([[sample_size, effect_size]])[0]

alpha_final = adjust_for_risk(alpha_base, risk_level)
alpha_final = adjust_for_tail(alpha_final, tail_type)
alpha_final = adjust_for_multiple_tests(alpha_final,
    num_tests)

alpha_final = max(0.001, min(alpha_final, 0.15))
```

The overall prediction workflow of the deployed system is illustrated in Figure 5.3.

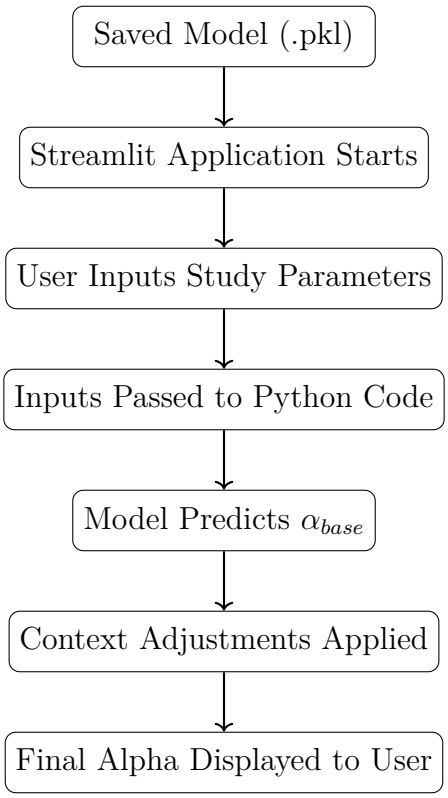

Figure 5.3.: Prediction workflow of the deployed system

5.2.3. Displaying Results to the User

After the calculations are completed, the final recommended alpha value is displayed clearly in the interface.

Figure 5.4 shows the interface after the user has entered the inputs and generated the recommendation. The highlighted output box presents the final context adjusted alpha value.

In the example shown, a study with a sample size of 241, an expected effect size of 0.30, a lenient risk tolerance, a one tailed test, and one hypothesis test results in a recommended alpha value of 0.0747. This value is generated through the predictive model and subsequent contextual adjustments.

After the α_{base} value has been computed, the application must communicate the result to the user through the interface. The following code implements the output component of the deployed application. It retrieves the final alpha recommendation generated by the prediction pipeline and displays it in the Streamlit interface. This step completes the interaction cycle by presenting the result of the model prediction and contextual adjustments in a clear and accessible format for the user.

Figure 5.4.: Interface of the system after entering the values

The display is implemented as follows:

```
st.write("Recommended significance level (alpha):")
st.success(round(alpha_final, 4))
```

This allows the user to immediately view the recommended alpha value based on the provided study parameters.

5.2.4. Running the Application

To allow other researchers to use the developed system, the implementation code will be made available through a public GitHub repository. This allows users to download the application and run it on their own computer.

To execute the system, the user first downloads or clones the repository from GitHub. The repository contains the trained model file, the application code, and the required dependencies.

After downloading the project, the required Python libraries must be installed. These include libraries such as streamlit, pandas, numpy, and scikit-learn.

Once the dependencies are installed, the application can be started using the following command in the project directory:

```
streamlit run app.py
```

When the application starts, Streamlit automatically opens the interface in a web browser. The user can then enter the required study parameters,

such as sample size, effect size, risk tolerance, test type, and number of hypothesis tests.

After providing the inputs and clicking the recommendation button, the system generates and displays the recommended alpha value.

This setup allows the Alpha Recommendation System to be easily executed and reused by other researchers.

The source code for the system is available in the project repository `https://github.com/Abishek700/AlphaPredictionSystem`.

Additional information about the repository and instructions for running the application are provided in AppendixA.8.

5.3. Summary

The deployed Streamlit application provides a complete and self contained tool for α recommendation. Users with no programming experience can obtain a context adjusted significance level by entering five study parameters through a simple interface. The following chapter evaluates whether the recommendations produced by this system would have supported more reliable statistical decisions compared to the conventional threshold of $\alpha = 0.05$.

Part VI.

Testing

6. Testing

6.1. Study Selection for Evaluation

This section describes the source of the studies used to evaluate the proposed alpha recommendation system and explains the criteria used for their selection.

The studies used in this thesis are drawn from the FORRT Replication Database (FReD), a publicly available database that compiles published replication studies across multiple research fields[Fra26]. These studies are used to examine whether the recommended alpha values generated by the system would have led to more reliable statistical decisions. Each entry in the database links an original study with its corresponding replication outcome and includes key study characteristics such as sample size and effect size.

The database contains studies from several disciplines, including psychology, behavioral science, social science, economics, and education. These fields were selected because null hypothesis significance testing is widely used and documented replication results are available.

For this evaluation, 20 studies were selected based on three criteria. First, the original study reported a statistically significant result using the conventional threshold of $\alpha = 0.05$. Second, an independent replication study was conducted. Third, the replication did not support the original finding. These cases are important because they show situations where a result was initially considered significant but later failed to replicate. By focusing on these studies, the evaluation examines whether a different choice of alpha, as recommended by the system, could have led to a more cautious and potentially more reliable decision.

The selected studies differ in sample size, effect size, and research field. This variation allows the system to be tested across different study settings rather than within a single discipline6.1. It should be noted that this evaluation uses a deliberately selected sample of failed replications. This design allows the system's recommendations to be assessed in cases where the conventional $\alpha = 0.05$ demonstrably led to unreliable decisions. However, it does not evaluate the system's behaviour on studies that successfully replicated, which would be necessary for a complete performance

assessment. This limitation is discussed further in Chapter 7.

Table 6.1.: Selected Studies and Recommended Alpha Values

Study	Field	Sample Size	Effect Size	Replication Outcome	Recommended Alpha
1	Psychology	33	0.43	Failed	0.0325
2	Psychology	33	0.62	Failed	0.063
3	Psychology	34	0.54	Failed	0.050
4	Psychology	35	0.61	Failed	0.060
5	Psychology	36	0.54	Failed	0.042
6	Behavioral Science	170	0.19	Failed	0.040
7	Behavioral Science	170	0.19	Failed	0.040
8	Behavioral Science	172	0.15	Failed	0.050
9	Behavioral Science	172	0.15	Failed	0.050
10	Behavioral Science	139	0.17	Failed	0.0537
11	Behavioral Science	424	0.10	Failed	0.020
12	Behavioral Science	424	0.11	Failed	0.020
13	Behavioral Science	398	0.12	Failed	0.030
14	Behavioral Science	398	0.18	Failed	0.020
15	Psychology	48	0.40	Failed	0.030
16	Psychology	168	0.25	Failed	0.040
17	Psychology	41	0.31	Failed	0.020
18	Psychology	85	0.25	Failed	0.040
19	Education	6	0.52	Failed	0.080
20	Education	744	0.27	Failed	0.001

6.2. Procedure for Testing

For testing we evaluate the developed alpha recommendation system by predicting the recommended alpha for the selected 20 studies.

All selected studies originally rejected the null hypothesis using the conventional significance level of $\alpha = 0.05$. However, these findings failed to replicate. These cases therefore represent situations in which the original statistical decision was likely a false positive.

For each study, the given inputs sample size and effect size were entered into the alpha recommendation system and the context based adjustments are set to default(A neutral risk tolerance, a two tailed test, and a single hypothesis test). Based on these inputs, the system generated recommended alpha value will be compared with the fixed threshold of $\alpha = 0.05$.

6.2.1. Decision Comparison

Table 6.1 also contains the recommended alpha coloumn for each studies which is nothing but output from the alpha recommendation system.

The recommended alpha values ranged from 0.001 to 0.08, demonstrating that the system adapts the significance level according to study characteristics rather than applying a single fixed threshold$\alpha = 0.05$.

Among the 20 studies, 11 received a recommended alpha value lower than 0.05, indicating a stricter decision rule. Three studies received a recommendation equal to 0.05, and six received a recommendation higher than 0.05.

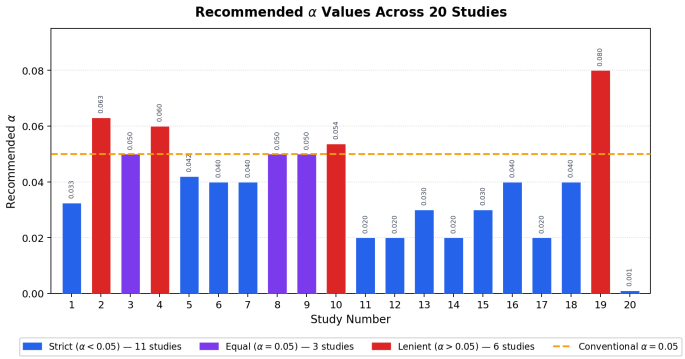

Figure 6.1.: Comparison of Recommended α Values Across 20 Studies

In the 11 cases where the recommended alpha was stricter than 0.05, the null hypothesis would not have been rejected under the proposed system. Because these studies later failed to replicate, these cases represent instances in which the system would have avoided potentially misleading rejections.

In the three cases where the recommended alpha matched 0.05, the system's recommendation was consistent with the conventional threshold.

In the six cases where a more lenient alpha was recommended, the system reflected a higher tolerance for false positives based on study characteristics. A more lenient α recommendation in a failed replication case represents an instance where the system, using only sample size and effect size under neutral contextual settings, did not identify the need for a stricter threshold. These cases suggest that the contextual adjustment inputs particularly risk tolerance may play an important role in refining recommendations for studies where the data driven baseline alone is insufficient. Providing accurate researcher preferences, rather than default neutral settings, could shift these recommendations toward stricter values.

6.2.2. Overall Result

Compared with the conventional use of $\alpha = 0.05$, the proposed alpha recommendation system provides improved decision support in several failed replication cases. As illustrated in Figure6.1, the blue bars represent studies for which the system recommends stricter significance levels ($\alpha < 0.05$), suggesting that a lower threshold could help reduce the risk of false positive conclusions. These stricter recommendations appear in than half of the evaluated studies, indicating that the conventional threshold may often be too lenient for the given study conditions. In three cases, the system retained the conventional threshold ($\alpha = 0.05$), indicating that the standard level was appropriate under those conditions. Some studies received more lenient recommendations ($\alpha > 0.05$); however, these values were generated using default contextual settings because the original studies did not report researcher preferences such as risk tolerance or testing context. Incorporating such contextual information would allow the adjustment stage of the system to better tailor the recommended α values to the specific study conditions.

Table 6.2.: Summary of Alpha Recommendation Outcomes

Category	Number of Studies	Interpretation
Stricter α (< 0.05)	11	Lower threshold suggested to reduce false positives
Equal α ($= 0.05$)	3	Conventional threshold considered appropriate
Lenient α (> 0.05)	6	Higher threshold suggested; may change with proper study context
Total Studies	20	–

Part VII.

Conclusion

7. Conclusion and Future Directions

7.1. Summary of the Study

This thesis addressed the problem of using a fixed significance level, commonly $\alpha = 0.05$, in hypothesis testing. Although widely used, this threshold is often applied without considering the specific characteristics of a study[CD82]. Such uniform use may contribute to misleading statistical decisions and replication failures[Ben+18][WL16].

To address this issue, a alpha recommendation system was developed. The system uses replication data to learn patterns between sample size, effect size, and study outcomes. A Random Forest regression model was trained to estimate a α_{base} value based on these study characteristics. In addition to this data driven component, contextual adjustments were incorporated to account for user preferences such as risk tolerance, test direction, and multiple hypothesis testing.

The system was implemented in Python and deployed through a Streamlit interface, allowing users to input study information and receive a recommended significance level in a structured and accessible manner.

7.2. Main Findings

The main finding is that the system recommended a significance level stricter than 0.05 in 11 out of 20 cases where the conventional threshold had led to a false positive decision, suggesting that a data driven approach can support more cautious and reproducible statistical conclusions.

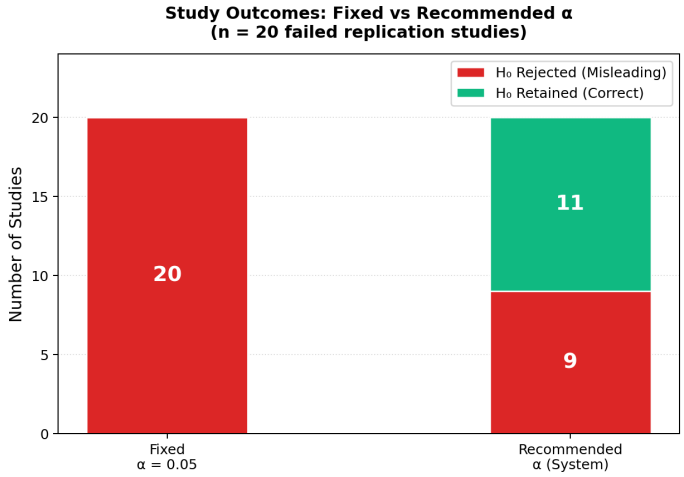

Figure 7.1.: Comparison of study outcomes

Overall, the findings support the idea that the significance level should not be treated as a universal constant but instead should reflect both empirical patterns from prior research and the specific context of each study.

7.3. Limitations

Several limitations of this study should be noted.

First, the evaluation was conducted using 20 failed replication studies from different research fields. Although these studies provide useful examples, a larger number of studies would allow a broader evaluation of the system.

Second, the contextual adjustment factors used in the system were implemented using heuristic scaling rules. These adjustments were inspired by ideas from decision theory[Ber85], but the exact numerical values were chosen for simplicity and interpretability.

Third, neutral contextual settings were used during the evaluation. This decision was made because the original studies did not report the researchers' preferences, such as their risk tolerance or testing strategy. Using neutral settings allowed the studies to be compared under the same conditions, but it does not fully capture how different researcher preferences might influence the recommended alpha values.

Fourth, the proposed system does not determine sample size and does not replace traditional power analysis. Instead, it assumes that important study parameters, such as sample size and expected effect size, are already specified by the researcher.

Overall, these limitations indicate that the proposed system should be viewed as a decision support tool that complements existing statistical practices rather than replacing established methods of study design and statistical inference.

7.4. Future Research Directions

Future research can extend the proposed system in several directions.

First, the contextual scaling factors used in this study could be further refined using formal decision theoretic models. Developing such models would require a deeper theoretical analysis and empirical calibration across many studies, which is beyond the scope of the present work.

Second, the machine learning model could be trained on a larger and more diverse collection of replication datasets from additional scientific disciplines. Expanding the dataset would allow the system to learn broader statistical patterns and improve the generalizability of the alpha recommendations.

Third, future studies could compare the proposed approach with alternative statistical frameworks, such as Bayesian methods [Ber85] or adaptive error control procedures[BH95]. Such comparisons would help evaluate how the proposed system performs relative to other approaches for managing statistical uncertainty.

Fourth, the system could be extended to integrate power analysis. This would allow the framework to provide guidance not only on the choice of significance level but also on appropriate sample size during study design.

Finally, future work could explore the development of field specific models that account for differences in research practices and replicability patterns across disciplines. This could help tailor alpha recommendations to the methodological characteristics of particular scientific fields.

7.5. Final Remarks

This thesis demonstrates that significance level selection can be approached as a data informed and context sensitive decision rather than as a fixed convention. By combining replication evidence with user defined preferences, the proposed alpha recommendation system provides a structured framework for more flexible statistical decision making.

While further refinement is possible, the study contributes to the ongoing discussion on improving research reliability and encourages a more

thoughtful approach to hypothesis testing across disciplines.

Bibliography

[BH95] Y. Benjamini and Y. Hochberg. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing". In: *Journal of the Royal Statistical Society: Series B* 57.1 (1995), pp. 289–300.

[Bak16] M. Baker. "1,500 Scientists Lift the Lid on Reproducibility". In: *Nature* 533.7604 (2016), pp. 452–454. DOI: 10.1038/533452a.

[Bec00] L. A. Becker. *Effect Size (ES)*. Accessed: 2026-02-25. University of Colorado Colorado Springs. 2000. URL: https://www.uv.es/~friasnav/EffectSizeBecker.pdf.

[Ben+18] D. J. Benjamin et al. "Redefine statistical significance". In: *Nature Human Behaviour* 2 (2018), pp. 6–10.

[Ber85] J. O. Berger. *Statistical Decision Theory and Bayesian Analysis*. New York: Springer, 1985. DOI: 10.1007/978-1-4757-4286-2.

[Bre01] L. Breiman. "Random Forests". In: *Machine Learning* 45.1 (2001), pp. 5–32. DOI: 10.1023/A:1010933404324. URL: https://www.stat.berkeley.edu/~breiman/randomforest2001.pdf.

[CD82] M. Cowles and C. Davis. "On the Origins of the .05 Level of Statistical Significance". In: *American Psychologist* 37.5 (1982), pp. 553–558. DOI: 10.1037/0003-066X.37.5.553.

[Cha25] J. Chang. *Pre-Hacking: Can Behavioral Experimenters Make Design Decisions To Get The Results They Want?* 2025. URL: https://osf.io/s9p7a.

[Coh88] J. Cohen. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.

[ED21] T. M. Errington and A. Denis. *Replication Data from the Reproducibility Project: Cancer Biology*. 2021. DOI: 10.17605/OSF.IO/E5NVR. URL: https://osf.io/e5nvr.

[FD17] E. D. Foster and A. Deardorff. "Open Science Framework (OSF)". In: *Journal of the Medical Library Association* 105.2 (2017), pp. 203–206. DOI: 10.5195/jmla.2017.88.

[Fis25] R. A. Fisher. *Statistical Methods for Research Workers*. Edinburgh: Oliver and Boyd, 1925. URL: https://archive.org/details/in.ernet.dli.2015.205971/mode/2up.

[Fra26] Framework for Open and Reproducible Research Training (FORRT). *FReD Explorer: FORRT Replication Database*. Accessed: 2026-03-07. 2026. URL: https://forrt.org/explorer/.

[HFC19] M. H. Herzog, G. Francis, and A. Clarke. "The Multiple Testing Problem". In: *Understanding Statistics and Experimental Design*. Cham: Springer, 2019, pp. 63–66. DOI: 10.1007/978-3-030-03499-3_5.

[Ha23] V. T. Ha. "Experimental Study on Remaining Useful Life Prediction of Lithium-Ion Batteries Based on Three Regression Models for Electric Vehicle Applications". In: *Preprints* (2023). DOI: 10.20944/preprints202306.0999.v1.

[Hal23] A. H. Hales. "One-Tailed Tests: Let's Do This (Responsibly)". In: *Psychological Methods* 29.6 (2023), pp. 1209–1218. DOI: 10.1037/met0000610.

[Kim15] J. H. Kim. *How to Choose the Level of Significance: A Pedagogical Note*. Tech. rep. 66373. Munich Personal RePEc Archive, 2015. URL: https://mpra.ub.uni-muenchen.de/66373/1/MPRA_paper_66373.pdf.

[Kim17] H.-Y. Kim. "Statistical notes for clinical researchers: Chi-squared test and Fisher's exact test". In: *Restorative Dentistry & Endodontics* 42.2 (2017), pp. 152–155. DOI: 10.5395/rde.2017.42.2.152. URL: https://rde.ac/journal/view.php?doi=10.5395/rde.2017.42.2.152.

[Kwa23] S. Kwak. "Are Only p-Values Less Than 0.05 Significant? A p-Value Greater Than 0.05 Is Also Significant!" In: *Journal of Lipid and Atherosclerosis* 12.2 (2023), pp. 89–95. DOI: 10.12997/jla.2023.12.2.89.

[Li21] X. Li. *Challenging the N-Myopia: Effect Size, Not Sample Size, Predicts the Replicability of Psychological Science*. 2021. URL: https://osf.io/5nvrt.

[ML22] M. Maier and D. Lakens. "Justify Your Alpha: A Primer on Two Practical Approaches". In: *Advances in Methods and Practices in Psychological Science* 5.2 (2022). DOI: 10.1177/25152459221080396. URL: https://doi.org/10.1177/25152459221080396.

[Mud+12] J. F. Mudge et al. "Setting an Optimal α That Minimizes Errors in Null Hypothesis Significance Tests". In: *PLOS ONE* 7.2 (2012), e32734. DOI: 10.1371/journal.pone.0032734. URL: https://doi.org/10.1371/journal.pone.0032734.

[NP33] J. Neyman and E. S. Pearson. "On the Problem of the Most Efficient Tests of Statistical Hypotheses". In: *Philosophical Transactions of the Royal Society of London. Series A* 231.694-706 (1933), pp. 289–337. DOI: 10.1098/rsta.1933.0009.

[Nos+22] B. A. Nosek et al. "Replicability, Robustness, and Reproducibility in Psychological Science". In: *Annual Review of Psychology* 73 (2022), pp. 719–748. DOI: 10.1146/annurev-psych-020821-114157.

[Ope15] Open Science Collaboration. "Estimating the Reproducibility of Psychological Science". In: *Science* 349.6251 (2015). DOI: 10.1126/science.aac4716.

[PF21] S. J. Peterson and S. Foley. "Clinician's Guide to Understanding Effect Size, Alpha Level, Power, and Sample Size". In: *Nutrition in Clinical Practice* 36.3 (2021), pp. 598–605. DOI: 10.1002/ncp.10674.

[RRS17] M. Ryo, M. C. Rillig, and B. Statzner. "Statistically Reinforced Machine Learning for Nonlinear Patterns and Variable Interactions". In: *Ecosphere* 8.11 (2017), e01976. DOI: 10.1002/ecs2.1976. URL: https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs2.1976.

[Ric11] J. T. Richardson. "Eta squared and partial eta squared as measures of effect size in educational research". In: *Educational Research Review* 6.2 (2011), pp. 135–147. ISSN: 1747-938X. DOI: https://doi.org/10.1016/j.edurev.2010.12.001. URL: https://www.sciencedirect.com/science/article/pii/S1747938X11000029.

[Sci23] Scikit-Learn Developers. *scikit-learn: Machine Learning in Python*. Accessed: 2026-03-03. scikit-learn. 2023. URL: https://scikit-learn.org/stable/.

[Sot17] C. J. Soto. *The Life Outcomes Of Personality Replication Project*. 2017. URL: https://osf.io/d3xb7.

[Sot22] L. K. Sotola. *Applying Z-Curve to Large-Scale Replication Studies*. 2022. URL: https://osf.io/uge5r.

[Str24] Streamlit. *Streamlit Documentation*. https://docs.streamlit.io/. Accessed: 2026-03-04. 2024.

[WL16] R. L. Wasserstein and N. A. Lazar. "The ASA's Statement on p-Values: Context, Process, and Purpose". In: *The American Statistician* 70.2 (2016), pp. 129–133. DOI: 10.1080/00031305.2016.1154108.

[WM05] C. J. Willmott and K. Matsuura. "Advantages of the Mean Absolute Error (MAE) over the Root Mean Square Error (RMSE) in Assessing Average Model Performance". In: *Climate Research* 30 (2005), pp. 79–82. DOI: 10.3354/cr030079. URL: https://www.int-res.com/abstracts/cr/v30/cr030079.

A. Supplementary Materials and Implementation Details

This appendix summarizes the key mathematical expressions and implementation details used in the Alpha Recommendation System. The formulas listed here provide a compact reference for the computational procedures described in Chapter 4.

A.1. Baseline Alpha Prediction

The baseline significance level is predicted using a Random Forest regression model trained on replication data.

$$\alpha_{\text{base}} = f(N, d) \tag{A.1}$$

where N represents the sample size and d denotes the expected standardized effect size.

A.2. Alpha Adjustment Pipeline

The baseline alpha predicted by the model is adjusted through a sequence of contextual transformations.

A.2.1. Risk Tolerance Adjustment

$$\alpha' = \alpha_{\text{base}} \cdot k_1 \tag{A.2}$$

where k_1 represents the risk tolerance scaling factor.

A.2.2. Test Direction Adjustment

$$\alpha'' = \begin{cases} 2\alpha' & \text{if one-tailed} \\ \alpha' & \text{if two-tailed} \end{cases} \tag{A.3}$$

This adjustment accounts for whether the statistical test is one-tailed or two-tailed.

A.2.3. Multiple Testing Correction

$$\alpha_{\text{final}} = \frac{\alpha''}{m} \tag{A.4}$$

where m denotes the number of simultaneous hypothesis tests.

A.3. Optimal Alpha Definition

For studies in which the replication does not confirm the original finding, a study-specific optimal significance level is derived using the original reported p-value.

$$\alpha_{\text{opt}} = \max\{\alpha \in A : \alpha < p_{\text{orig}}\} \tag{A.5}$$

This rule selects the largest candidate alpha value that remains strictly smaller than the reported p-value.

A.4. Candidate Alpha Set

The candidate set of significance levels used for selecting the optimal alpha is defined as

$$A = \{0.001, 0.005, 0.010, 0.020, 0.030, 0.040, 0.050, 0.060, 0.090, 0.100\} \tag{A.6}$$

Using a discrete set of candidate values ensures that the resulting thresholds remain interpretable and comparable across studies.

A.5. Statistical Recovery Formulas

When original studies did not explicitly report sample size or effect size, these quantities were reconstructed using standard statistical relationships derived from reported test statistics.

A.5.1. Effect Size from t-test

$$r = \sqrt{\frac{t^2}{t^2 + df}} \tag{A.7}$$

A.5.2. Sample Size from Degrees of Freedom

$$N = df + 2 \tag{A.8}$$

A.5.3. Effect Size from Chi-Squared Test

$$\phi = \sqrt{\frac{\chi^2}{N}} \tag{A.9}$$

A.5.4. Effect Size from F-test

$$\eta^2 = \frac{F \cdot df_1}{F \cdot df_1 + df_2} \tag{A.10}$$

A.5.5. Conversion from η^2 to Correlation

$$r = \sqrt{\eta^2} \tag{A.11}$$

A.6. Model Evaluation Metric

The performance of the Random Forest regression model is evaluated using the Mean Absolute Error (MAE).

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{A.12}$$

where y_i represents the observed optimal alpha values and $\hat{y}_i$ denotes the predicted values.

A.7. Python Implementation

The Alpha Recommendation System was implemented in Python using the `scikit-learn` machine learning library.

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor

data = pd.read_excel("master_alpha_dataset_updated.xlsx")

X = data[["sample_size","effect_size"]]
y = data["best_alpha"]

model = RandomForestRegressor(n_estimators=200, random_state
    =42)
model.fit(X, y)
```

A.8. Source Code Repository

The full implementation of the Alpha Recommendation System, including the trained model, dataset, and Streamlit application, is available in the public GitHub repository:

`https://github.com/Abishek700/AlphaPredictionSystem`

The repository contains the dataset used for training, the Python scripts for model training and evaluation, the serialized machine learning model, and the Streamlit application used to generate alpha recommendations.

A.8.1. Running the Application Locally

```
# 1. Clone the repository
git clone https://github.com/Abishek700/
    AlphaPredictionSystem
cd AlphaPredictionSystem

# 2. Install dependencies
pip install -r requirements.txt

# 3. Launch the application
streamlit run app.py
```

Once launched, Streamlit opens the interface automatically in the default web browser, allowing users to enter study parameters and generate alpha recommendations.